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1. Define modulation and explain its purpose in wireless communication.

Answer: Modulation is the process of varying a carrier signal's properties, such as amplitude, frequency, or phase, to encode information for transmission. This process is essential for efficient use of the electromagnetic spectrum and minimizing interference between channels.

Score: 7.0/10

Justification: The student's answer demonstrates a good understanding of the definition and importance of modulation, but lacks depth and application in their explanation. The answer is clear and well-organized, but does not provide specific examples or explanations of how modulation helps to achieve its goals, nor does it provide a detailed explanation of the modulation process or its underlying principles.

Feedback: To improve, the student should provide specific examples or explanations of how modulation helps to achieve its goals, choose a specific wireless communication scenario and explain how modulation is used in that scenario, and provide a more detailed explanation of the modulation process, including the different types of modulation and how they work.

2. Calculate the path loss at a distance of 500 meters for a signal frequency of 2 GHz in free space.

Answer: The FSLP formula is $dB = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55$

$$\text{FSLP(dB)} = 20 \log_{10}(500) + 20 \log_{10}(2) - 147.55$$

$$= 20 \cdot 2.69897 + 20 \cdot 9.30103 - 147.55$$

$$= 53.98 + 186.02 - 147.55$$

$$= 92.45 \text{ dB}$$

Score: 10.0/10

Justification: The student correctly calculates the path loss using the Free-Space Path Loss formula, but lacks analysis and explanation of the result. The answer is clear, but formatting and notation could be improved.

Feedback: The student is encouraged to provide a brief analysis of the result, explaining its significance and implications in the context of wireless communication. Additionally, proper mathematical notation and formatting are suggested to make the answer easier to read and understand.

Summary:

Total Score: 17.0

Percentage: 34.00%

Grade: F

Overall Feedback: To improve, the student should provide specific examples or explanations of how modulation helps to achieve its goals, choose a specific wireless communication scenario and explain how modulation is used in that scenario, and provide a more detailed explanation of the modulation process, including the different types of modulation and how they work. Additionally, a brief analysis of the result, explaining its significance and implications in the context of wireless communication, is encouraged. Proper mathematical notation and formatting are also suggested to make the answer easier to read and understand.